

Observation Attenuation for Repeated-Use Privacy Budgets: Mixture/Subsampling Bounds and a Stub-Mix Interface (Series Synthesis)

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Abstract

Several “Boss Fight” calculations in this series take an explicit *observation model*: how likely is it that an adversary sees a particular hop, contact set, or transcript projection. This note centralizes a small but repeatedly used fact: *partial observation is a mixture/subsampling channel that strictly attenuates MaxL/odds-inflation, and receipts should publish the attenuation parameter*. We give compact bounds (in the exact form needed by dial-sheet tables) and record a deployable interface: a *stub-mix* / *relay* that reduces observation probability without attempting full mixnet anonymity.

Scope. This is a short interface note. It does *not* claim that a stub mix alone solves reader privacy; it only formalizes how *reduced observation* feeds into the series’ accounting.

Imports. Endpoint metrics (odds inflation / MaxL) and their conversion rules are imported from the endpoint bridge.[1] Repeated-use accounting and the role of p_{obs} in dial sheets are imported from the Boss Fight series.[2] Max-mixture separations under deadlines are treated as owned by the stop-time addendum; we use only the simplest mixture algebra here.[3]

Exports. A tiny lemma suite for *observation attenuation* (mixtures/subsampling) and a concrete receipt-facing parameterization: publish p_{obs} (or a monitored upper bound) for each transcript projection tier, using the series line-item schema (Synthesis 12). For multi-tier combination rules, see the tiered-observation vectors note (Synthesis 15).[7, 6]

1 Observation as a channel wrapper

Many witness surfaces in this archive are not directly exposed to an adversary. For example: only a corrupted hop sees the next-hop choice; only a surveilled vantage sees packet times with good resolution; only a colluding relay learns contact sets. We package this as a wrapper channel.

Definition 1 (Observation thinning). Let Y be a transcript (or transcript projection) generated by a mechanism under secret C . Fix an “observation” probability $p \in [0, 1]$ independent of C . Define the *thinned* observation Z by

$$Z := \begin{cases} Y, & \text{with probability } p, \\ \perp, & \text{with probability } 1 - p, \end{cases}$$

where $\perp$ denotes “not observed” (or an uninformative placeholder).

This is the simplest model. More detailed models (multiple tiers, partial fields observed, noisy timestamps) reduce to mixtures of such wrappers.

2 MaxL and odds inflation contract under thinning

Maximal leakage admits a direct bound under thinning that is convenient for receipts.

Lemma 1 (Mixture form for MaxL under thinning). *Let Y be a channel from C with maximal leakage $\mathcal{L}(C \rightarrow Y) = b$. Let Z be the thinned observation with parameter p as above. Then*

$$\mathcal{L}(C \rightarrow Z) \leq \log_2((1-p) + p 2^b).$$

Proof. Maximal leakage equals $\log_2(\sum_z \max_c P(Z = z \mid C = c))$. For $z = \perp$, the maximum conditional probability is $1-p$. For $z \neq \perp$, $P(Z = z \mid c) = p P(Y = z \mid c)$, so the contribution is scaled by p . Thus $\sum_z \max_c P(Z = z \mid c) = (1-p) + p \sum_y \max_c P(Y = y \mid c) \leq (1-p) + p 2^b$. $\square$

Remark 1 (Dial-sheet interpretation). *If b is the per-lookup budget for a projection, the thinned budget is approximately pb for small b (first-order), but Lemma 1 gives the exact log-sum-exp form used by receipts.*

A companion inequality holds at the TV level.

Lemma 2 (TV contraction under thinning). *For any two secrets c, c' , let P_c and $P_{c'}$ denote the induced laws of Y . Let Q_c and $Q_{c'}$ denote the induced laws of Z . Then*

$$\text{TV}(Q_c, Q_{c'}) = p \text{TV}(P_c, P_{c'}).$$

Proof. $Q_c = (1-p)\delta_\perp + p P_c$ and similarly for c' . TV is homogeneous for signed measures and $\delta_\perp$ cancels. $\square$

3 A stub-mix / relay as an observation attenuator

The Boss Fight calculations show that dummy-only mechanisms can collapse unless p_{obs} is extremely small.[2] This motivates a *cheap* architectural lever: reduce observation probability without attempting full low-latency anonymity.

Interface. A *stub mix* is a one-hop (or few-hop) relay layer between the client and the DHT contact set. The goal is not to hide the query from the relay (that is the job of cryptographic query privacy), but to reduce the probability that a routing-layer adversary links the client to the contact set for any given lookup. Operationally this can be instantiated as:

- an OHTTP-style relay for lookup messages,
- a small anycast/broker set with rotating, audited egress,
- a short mix/cover network used only for fallback lookups.

Receipt parameterization. Receipts should treat the stub mix as a change to the observation model: replace p_{obs} for the fallback path with a (published) upper bound that reflects relay compromise, egress observation, and correlation risk. Lemma 1 then converts this attenuation into a bound on the budget actually spent.

Remark 2 (Honesty requirement). *A stub-mix design is only useful if p_{obs} is published in the same ABOM/OINL or release receipt that publishes the Boss-Fight budget (as a line item).[7] Otherwise “privacy” can silently degrade when the relay set centralizes or becomes observed. Closed-view auditing and CPFC receipts are a natural monitoring hook for this parameter.[4]*

4 How this composes with the tiered pattern

The tiered primary/fallback pattern note recommends treating the raw DHT as a budgeted residual.[5] Observation attenuation is orthogonal: it can be applied to either path, but is particularly valuable for the fallback because it directly loosens the dummy-cover pressure.

References

- [1] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Series synthesis note (Synthesis 5), 2026. (This archive.)
- [2] Anonymous. AnonDHT in the Boss Fight Regime: Compiling a MaxL Budget into Lookup Knobs. Boss Fight series Paper B, 2026. (This archive.)
- [3] Anonymous. Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms. Published backbone note (2026-01-26). Wiki: `[[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]]`.
- [4] Anonymous. Closed-View Auditing and CPPC Manifests. AnonDHT state series Paper 3, 2026. (This archive.)
- [5] Anonymous. A Tiered Deployment Pattern for Reader Privacy: Primary Cryptographic Path, Budgeted DHT Fallback. Series synthesis note (Synthesis 10), 2026. (This archive.)
- [6] Anonymous. Tiered Observation Vectors for Anonymous-DHT Receipts: Contact-Set, Timing, and Congestion Tiers with Conservative MaxL Composition. Synthesis 15, The-One-True-Archive (2026).
- [7] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Series synthesis note (Synthesis 12), The-One-True-Archive (2026).